Glen Muthoka Mutinda

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Professional Summary

Final-year BEng Electrical & Electronics Engineering student at the University of Southampton, with hands-on experience in digital IC physical implementation, FPGA RTL and CPU microarchitecture. Led a TSMC 65nm ring oscillator IC from schematic through custom layout, Mentor Calibre DRC/LVS sign-off and GDSII submission. Seeking a graduate CPU physical implementation role spanning RTL through place-and-route, sign-off and low-power optimisation.

Education

University of Southampton **Southampton, UK**
BEng Electrical & Electronics Engineering, Expected First Class Honours *2023–2026*
Modules: Digital Systems Design, Integrated Circuit Design, Embedded Systems, Signal Processing, Control Systems, Network Security. **Year 2 Project:** TSMC 65nm Ring Oscillator IC (Team Lead). **Final Year Project:** AI-Driven Optical Authentication using Optical PUFs.
2022–2023: Engineering Foundation Programme – Distinction (ONCAMPUS Global, Southampton).

Relevant Experience

ARTEMIS Small Sat-1 Lunar CubeSat Project **University of Southampton**
Junior Software Engineer *Sep 2023–Jun 2025*

- Developed and debugged C/C++ flight-software components for an 8-engineer student lunar CubeSat across a 2-year mission cycle, operating within MISRA C and NASA-style coding constraints.
- Implemented real-time telemetry processing under strict timing budgets, contributed on-board sensor-interfacing modules, and built unit-test harnesses to improve integration confidence.
- Operated daily in Unix-based build and Git version-control environments, reinforcing interface discipline across hardware, software and systems sub-teams.

Selected Hardware & Implementation Projects

Sep 2023–Jun 2024: **TSMC 65nm Real-Time Clock: Ring Oscillator IC** – Team Lead, 6-engineer team
Cadence S-Edit, T-Spice, L-Edit; Mentor Calibre DRC/LVS

- Led a 6-engineer team through the full schematic-to-GDSII physical implementation flow on a TSMC 65nm process, delivering a ring oscillator block for a real-time clock IC.
- Designed the transistor-level schematic in S-Edit, ran T-Spice simulations for frequency stability and timing closure, and implemented custom L-Edit layout with nanometre-aware floorplanning.
- Cleared Mentor Calibre DRC and LVS sign-off with zero violations, submitted GDSII for tape-out, and documented the methodology for the following cohort.

Apr 2025–Jun 2025: **MIPSquare++: 5-Stage Pipelined MIPS Processor Simulator** Bananz0/MIPSquare
C++23, CMake, Git

- Built a cycle-accurate C++23 simulator of a 5-stage MIPS pipeline (Fetch, Decode, Execute, Memory, Write-Back), demonstrating CPU microarchitecture fluency relevant to RTL collaboration.
- Implemented data forwarding between EX/MEM and MEM/WB stages, load-use hazard detection with automatic stalling, and branch flushing to approach 1 CPI on standard test programs.
- Engineered a MIPS assembly parser for R-, I- and J-type instructions plus register-file and instruction/data memory models, validated under Unix-based CMake builds.

Nov 2024–Dec 2024: **16-Stage FIR Notch Filter for Real-Time Audio on FPGA** Bananz0/FIR-Notch-Filter
SystemVerilog, Quartus Prime, ModelSim, MATLAB; Altera Cyclone V DE1-SoC

- Designed a parameterised 16-stage FIR notch filter in SystemVerilog with a 4-state FSM datapath/controller, achieving 19-cycle latency at 50 MHz on Altera Cyclone V.
- Generated and validated filter coefficients in MATLAB, then verified RTL functionality in ModelSim before Quartus Prime synthesis and timing-aware FPGA implementation.
- Synthesised to 2,847 logic elements (approximately 5% FPGA utilisation) and processed dual-channel 48 kHz stereo audio with -40 dB notch depth.

Jan 2025–Mar 2025: **WattsApp: Embedded Smart System Management Platform** Bananz0/WattsApp
Embedded C, UART, I2C, ADC, InfluxDB, Grafana; AVR ATMega644p

- Built bare-metal embedded C firmware on an AVR ATMega644p, using interrupt-driven sensor polling and UART telemetry under Unix-based toolchains.
- Implemented I2C and ADC interfaces for live current and voltage acquisition, integrated with an InfluxDB time-series store and Grafana dashboard for real-time visibility.
- Profiled and optimised the main loop to keep critical-path response within sub-millisecond bounds, demonstrating low-level hardware/software debugging.

Sep 2025–Jun 2026: **QuantumID: Optical PUF Authentication Framework** – Lead Developer & Researcher, Final Year Project
Python, PyTorch, NumPy, SciPy, scikit-learn

- Developed an Optical Physical Unclonable Function authentication system in Python using excitation-emission matrix fluorescence spectra and a custom convolutional autoencoder.
- Engineered a physics-aware preprocessing and augmentation pipeline, then trained reconstruction and contrastive-learning stages to produce latent fingerprints and binary cryptographic hashes.
- Automated the experiment workflow with NumPy and SciPy under Unix tooling, applying data-driven verification methodology relevant to silicon characterisation flows.

Technical Skills

Physical Impl. & EDA: TSMC 65nm process exposure, RTL-to-GDSII flow, custom layout, DRC, LVS, place-and-route concepts, timing/area/power trade-offs, nanometre-process awareness, low-power design; Cadence S-Edit/T-Spice/L-Edit, Mentor Calibre, Quartus Prime, ModelSim, Vivado, LTSpice, MATLAB/Simulink.
HDL & Programming: SystemVerilog, Verilog, VHDL, RTL/FSM design, pipelined CPU microarchitecture, FPGA synthesis; C, C++, Python, MATLAB, Tcl (basic), Bash, PowerShell, Unix tools, Git, CMake; bare-metal C on AVR/STM32/ESP32, UART/I2C/SPI/ADC.